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Product-oriented Engineer / Product Builder

Product-oriented systems engineer focused on turning complex, ambiguous problem spaces into coherent, usable software systems. My work sits at the intersection of system architecture, execution models, and UX-driven product design, with hands-on experience taking ideas from research prototypes to user-facing platforms.

EDUCATION

Rikkyo University, Tokyo

B.A. in History (Expected Sep 2027)

Focus: institutional systems, bureaucracy, legal structures, and information organization

Supplemented with AI ethics, Information Science coursework

University Coursework(U.S.)

Virginia Tech / Chicago

Calculus I–II, Linear Algebra, Data structures & object-oriented programming

Humanities & design coursework

CORE SKILLS (Selected)

Product & System Design

- Product architecture & abstraction
- UX-driven system design
- Interaction model design
- Workflow & requirement structuring
- Tooling & prototype development

Engineering

- Languages: Python, TypeScript/JavaScript, C
- Frontend: React, TypeScript, Tailwind, Radix UI
- Backend: Node.js, Express, REST APIs, WebSocket
- Database: PostgreSQL, Drizzle ORM, Firebase Auth

AI & Applied Systems

- LLM API integration (OpenAI, Claude)
- Prompt architecture & symbolic systems
- Custom DSL design & C-based interpreter
- AI-assisted workflows (non-algorithm focus)

Design

- Figma, Adobe Suite, CSP
- UI system & interaction prototyping

SELECTED PROJECTS

Spiral — Language & Execution System

Founder & Lead Developer | 2024–Present

Designed and built an experimental system for managing complex event histories while preserving explainability and user-facing clarity.

- Designed a custom execution model and DSL to represent structured intent instead of raw commands.
- Built a traceable, append-only event-history kernel that computes a meaningful “recent state” from causal relationships rather than destructive updates.
- Took the system from research concepts to working prototypes with real users and UI constraints.
- Developed UX concepts for navigating structured event histories and AI-assisted workflows in a way that remains legible to end users.

spiral-core-series — Event History Kernel

Open-source research prototype | 2025

- Implemented an append-only event DAG with a computed "RECENT_A" frontier (v0.46), providing a causally explainable recent view without log truncation.
- Enforced strong invariants: no mutation, parent consistency, and conflict signals attributable to concrete input pairs.
- Used as the formal and technical foundation for the Spiral system.

Spiral.ooo — Product Prototype Platform

Developer & System Architect | 2025

- Built a full-stack platform for interacting with the Spiral system through a human-readable editor.
- Designed end-to-end product stack (React / Node / PostgreSQL / Stripe / AI APIs).
- Focused on usability, system coherence, and iterative product design.

Applied Systems & Internal Tools (Selected)

- Built event-driven automation and messaging systems in Python for QQ and Discord, focusing on reliability, state handling, and long-running workflows.
- Developed a research-oriented publication platform with structured citation management and AI-assisted summarization.
- Designed internal tools prioritizing workflow clarity, maintainability, and user-facing simplicity.

LANGUAGES

- Chinese (Native)
- English (Fluent)
- Japanese (Fluent / N1)